Chapter 2: Analytic Functions

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October 18, 2026

Outline

1. Concept of Analytic Functions

2. Necessary and Sufficient Conditions for Analyticity

3. Elementary Functions

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1. Concept of Analytic Functions

2. Necessary and Sufficient Conditions for Analyticity

3. Elementary Functions

1 Derivatives of Complex Functions

1) Definition of the Derivative

Definition

Let $w = f(z)$ be defined in a domain D, and let $z_0 \in D$ and $z_0 + \Delta z \in D$. If the limit

$$\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

exists (and is finite), then $f(z)$ is said to be differentiable (可导) at z_0.

Definition (Cont.)

This limit is called the derivative (導数) *of* $f(z)$ *at* z_0 *and is denoted by*

$$f'(z_0) = \left.\frac{\mathrm{d}w}{\mathrm{d}z}\right|_{z=z_0} = \lim_{\Delta z \to 0} \frac{f(z_0+\Delta z) - f(z_0)}{\Delta z}. \tag{2.1.1}$$

Equivalently, for every $\varepsilon > 0$ *there is a* $\delta(\varepsilon) > 0$ *such that* $0 < |\Delta z| < \delta$ *implies*

$$\left| \frac{f(z_0+\Delta z) - f(z_0)}{\Delta z} - f'(z_0) \right| < \varepsilon.$$

As for limits in Chapter 1, $\Delta z \to 0$ *may happen along any path.*

Example 1.1

Determine whether the following functions are differentiable:

1. $f(z) = z^2$;
2. $f(z) = 2x + yi$.

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Solution.

1. For every z in the complex plane,

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{(z+\Delta z)^2 - z^2}{\Delta z}$$
$$= \lim_{\Delta z \to 0} (2z + \Delta z) = 2z.$$

Hence $f(z) = z^2$ is differentiable everywhere, and $f'(z) = 2z$.

Solution (Cont.)

2 With $\Delta z = \Delta x + i\Delta y$,

$$\frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{2(x+\Delta x) + (y+\Delta y)i - (2x+yi)}{\Delta x + i\Delta y}$$
$$= \frac{2\Delta x + i\Delta y}{\Delta x + i\Delta y}.$$

If $z+\Delta z \to z$ along a line parallel to the real axis (Figure 2.1), then $\Delta y = 0$ and the quotient equals $\dfrac{2\Delta x}{\Delta x} = 2$.
If $z+\Delta z \to z$ along a line parallel to the imaginary axis, then $\Delta x = 0$ and the quotient equals $\dfrac{i\Delta y}{i\Delta y} = 1$.
The two limits differ, so $f(z) = 2x+yi$ is differentiable nowhere.

Solution (Cont.)

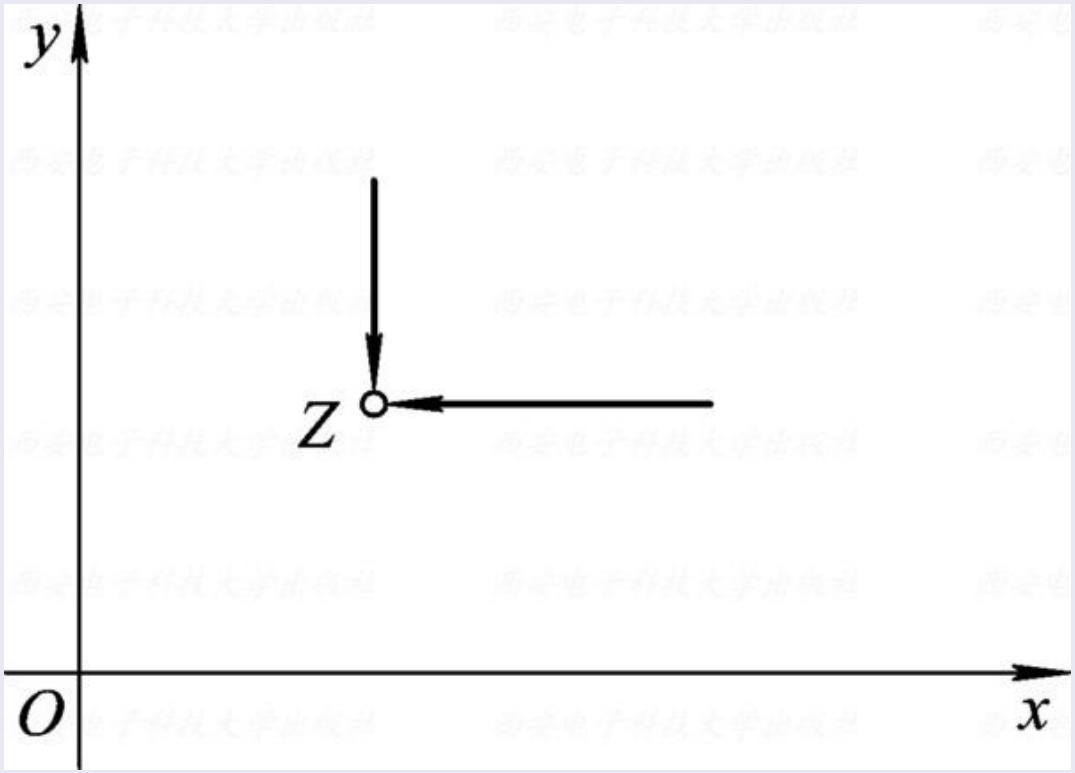


Figure: 2.1

2) Differentiability and Continuity
Suppose $f(z)$ is differentiable at z_0. Let

$$\rho(\Delta z) = \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} - f'(z_0). \tag{2.1.2}$$

By the definition of the derivative,

$$\lim_{\Delta z \to 0} \rho(\Delta z) = 0.$$

From (2.1.2),

$$f(z_0 + \Delta z) = f(z_0) + f'(z_0)\Delta z + \rho(\Delta z)\Delta z, \qquad (2.1.3)$$

and therefore

$$\lim_{\Delta z \to 0} f(z_0 + \Delta z) = f(z_0).$$

Thus, differentiability at z_0 implies continuity at z_0.
The converse is false: $f(z) = 2x + yi$ is continuous everywhere but differentiable nowhere.

3) Rules of Differentiation
Complex derivatives obey the same rules as real derivatives:

a) $(c)' = 0$, where c is a complex constant;
b) $(z^n)' = nz^{n-1}$, where n is a positive integer;
c) $[f(z) \pm g(z)]' = f'(z) \pm g'(z)$;
d) $[f(z)g(z)]' = f'(z)g(z) + f(z)g'(z)$;
e) $\left[\frac{f(z)}{g(z)}\right]' = \frac{f'(z)g(z) - f(z)g'(z)}{g^2(z)}$, where $g(z) \neq 0$;
f) $\{f[g(z)]\}' = f'(w)g'(z)$, where $w = g(z)$ (chain rule);
g) $f'(z) = \frac{1}{\varphi'(w)}$, where $w = f(z)$ and $z = \varphi(w)$ are mutually inverse single-valued functions and $\varphi'(w) \neq 0$.

4) Differential

As in real analysis, suppose $w = f(z)$ is differentiable at z_0. By (2.1.3),

$$\Delta w = f(z_0 + \Delta z) - f(z_0) = f'(z_0)\Delta z + \rho(\Delta z)\Delta z,$$

where $\lim_{\Delta z \to 0} \rho(\Delta z) = 0$. Hence $\rho(\Delta z)\Delta z$ is an infinitesimal of higher order than Δz, and $f'(z_0)\Delta z$ is the linear principal part of Δw.

The linear part $f'(z_0)\Delta z$ is called the differential (微分) of $w = f(z)$ at z_0:

$$\mathrm{d}w = f'(z_0)\Delta z. \tag{2.1.4}$$

For $f(z) = z$ we have $\mathrm{d}z = \Delta z$, so (2.1.4) becomes $\mathrm{d}w = f'(z_0)\,\mathrm{d}z$, that is,

$$f'(z_0) = \left.\frac{\mathrm{d}w}{\mathrm{d}z}\right|_{z=z_0}. \tag{2.1.5}$$

Thus, a function has a differential at z_0 if and only if it has a derivative at z_0. If $f(z)$ is differentiable at every point of a domain D, it is said to be differentiable in D.

2 Analytic Functions

Definition

If $f(z)$ is differentiable at z_0 and at every point of some neighborhood of z_0, then $f(z)$ is said to be analytic (解析) *at z_0.*
If $f(z)$ is analytic at every point of a domain D, then $f(z)$ is said to be analytic in D, or an analytic function in D.
If $f(z)$ is not analytic at z_0, then z_0 is called a singular point (奇点) *of $f(z)$.*

Analyticity is a neighborhood property: differentiability at a single point is not enough.

Example 1.2

Investigate the analyticity of the following functions:

1. $f(z) = z^2$;
2. $f(z) = 2x + yi$;
3. $f(z) = |z|^2$;
4. $f(z) = 1/z$.

Example 1.2

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1. $f(z) = z^2$;
2. $f(z) = 2x + yi$;
3. $f(z) = |z|^2$;
4. $f(z) = 1/z$.

Solution.

(1)(2) By Example 1.1, $f(z) = z^2$ is analytic in the whole plane, while $f(z) = 2x + yi$ is analytic nowhere.

(4) $f(z) = 1/z$ is differentiable for $z \neq 0$ with $f'(z) = -1/z^2$, so it is analytic in $\mathbb{C} \setminus \{0\}$, and $z = 0$ is a singular point.

Solution (Cont.)

(3) For $f(z) = |z|^2 = z\bar{z}$,

$$\frac{f(z_0+\Delta z)-f(z_0)}{\Delta z} = \frac{(z_0+\Delta z)(\overline{z_0}+\overline{\Delta z})-z_0\overline{z_0}}{\Delta z}$$
$$= \overline{z_0}+\overline{\Delta z}+z_0\frac{\overline{\Delta z}}{\Delta z}.$$

If $z_0 = 0$, the quotient equals $\overline{\Delta z} \to 0$, so $f'(0) = 0$.

Solution (Cont.)

If $z_0 \neq 0$, let $z_0 + \Delta z \to z_0$ along the line $y - y_0 = k(x - x_0)$. Then

$$\frac{\overline{\Delta z}}{\Delta z} = \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = \frac{1 - ik}{1 + ik},$$

which depends on k, so the limit does not exist.
Therefore, $f(z) = |z|^2$ is differentiable only at $z = 0$ and is analytic nowhere.

Theorem 1.1

1. *If $f(z)$ and $g(z)$ are analytic in a domain D, then $f \pm g$, fg, and f/g (where $g(z) \neq 0$) are analytic in D.*
2. *Suppose $h = g(z)$ is analytic in a domain D of the z-plane, $w = f(h)$ is analytic in a domain G of the h-plane, and $g(z) \in G$ for every $z \in D$. Then the composite function $w = f[g(z)]$ is analytic in D.*

Consequently, every polynomial is analytic in the whole plane, and every rational function $P(z)/Q(z)$ is analytic except at the zeros of $Q(z)$.

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1 Main Theorem

Theorem 2.1

Let $f(z) = u(x, y) + iv(x, y)$ be defined in a domain D. Then $f(z)$ is differentiable at a point $z = x + iy$ of D if and only if $u(x, y)$ and $v(x, y)$ are differentiable at (x, y) and satisfy the Cauchy–Riemann equations (柯西–黎曼方程)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

at this point.

Proof.

1. Necessity. Suppose $f(z)$ is differentiable at $z = x + iy$. By (2.1.3), for small $|\Delta z| = |\Delta x + i\Delta y| > 0$,

$$f(z+\Delta z) - f(z) = f'(z)\Delta z + \rho(\Delta z)\Delta z,$$

where $\lim_{\Delta z \to 0} \rho(\Delta z) = 0$. Write

$$f(z+\Delta z)-f(z) = \Delta u + i\Delta v, \quad f'(z) = a+ib, \quad \rho(\Delta z) = \rho_1 + i\rho_2.$$

Proof (Cont.)

Then

$$\begin{aligned}\Delta u + i\Delta v &= (a+ib)(\Delta x + i\Delta y) + (\rho_1 + i\rho_2)(\Delta x + i\Delta y) \\ &= (a\Delta x - b\Delta y + \rho_1\Delta x - \rho_2\Delta y) \\ &\quad + i(b\Delta x + a\Delta y + \rho_2\Delta x + \rho_1\Delta y).\end{aligned}$$

Comparing real and imaginary parts,

$$\begin{aligned}\Delta u &= a\Delta x - b\Delta y + \rho_1\Delta x - \rho_2\Delta y, \\ \Delta v &= b\Delta x + a\Delta y + \rho_2\Delta x + \rho_1\Delta y.\end{aligned}$$

Proof (Cont.)

Since $\rho_1, \rho_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$, u and v are differentiable at (x, y), with

$$\frac{\partial u}{\partial x} = a, \quad \frac{\partial u}{\partial y} = -b, \quad \frac{\partial v}{\partial x} = b, \quad \frac{\partial v}{\partial y} = a.$$

Hence

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Technical Note: Equivalent Forms of the Remainder

Let

$$r = \sqrt{(\Delta x)^2 + (\Delta y)^2}, \qquad R(\Delta x, \Delta y) = o(r).$$

Then the remainder（余项）can be written as

$$R = \varepsilon_1 \Delta x + \varepsilon_2 \Delta y, \qquad \varepsilon_1, \varepsilon_2 \to 0.$$

For $r > 0$, choose the following values, and set both to 0 at $r = 0$:

$$\varepsilon_1 = \frac{R\Delta x}{r^2}, \qquad \varepsilon_2 = \frac{R\Delta y}{r^2}.$$

Then $\varepsilon_1 \Delta x + \varepsilon_2 \Delta y = R$, and

$$|\varepsilon_1|, |\varepsilon_2| \leq \frac{|R|}{r} \to 0.$$

Proof (Cont.)

2. Sufficiency. Suppose u and v are differentiable at (x, y) and satisfy the Cauchy–Riemann equations. Write

$$f(z + \Delta z) - f(z) = \Delta u + i\Delta v.$$

Proof (Cont.)

By the preceding technical note, differentiability of u and v gives

$$\Delta u = \frac{\partial u}{\partial x}\Delta x + \frac{\partial u}{\partial y}\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y,$$
$$\Delta v = \frac{\partial v}{\partial x}\Delta x + \frac{\partial v}{\partial y}\Delta y + \varepsilon_3 \Delta x + \varepsilon_4 \Delta y,$$

where $\varepsilon_k \to 0$ $(k = 1, 2, 3, 4)$ as $(\Delta x, \Delta y) \to (0, 0)$. Therefore,

$$f(z + \Delta z) - f(z) = \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)\Delta x + \left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right)\Delta y + (\varepsilon_1 + i\varepsilon_3)\Delta x + (\varepsilon_2 + i\varepsilon_4)\Delta y.$$

Proof (Cont.)

By the Cauchy–Riemann equations,

$$\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y} = -\frac{\partial v}{\partial x} + i\frac{\partial u}{\partial x} = i\left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right).$$

Hence

$$\frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + (\varepsilon_1 + i\varepsilon_3)\frac{\Delta x}{\Delta z} + (\varepsilon_2 + i\varepsilon_4)\frac{\Delta y}{\Delta z}.$$

Proof (Cont.)

Since $\left|\frac{\Delta x}{\Delta z}\right| \leq 1$ and $\left|\frac{\Delta y}{\Delta z}\right| \leq 1$,

$$\lim_{\Delta z \to 0} \left[(\varepsilon_1 + i\varepsilon_3)\frac{\Delta x}{\Delta z} + (\varepsilon_2 + i\varepsilon_4)\frac{\Delta y}{\Delta z} \right] = 0.$$

Therefore,

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x},$$

so $f(z)$ is differentiable at $z = x + iy$.



By the proof of Theorem 2.1, if $f(z) = u + iv$ is differentiable at $z = x + iy$, then

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}. \tag{2.2.1}$$

Theorem 2.2

A function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D if and only if $u(x, y)$ and $v(x, y)$ are differentiable in D and satisfy the Cauchy–Riemann equations in D.

How to show that a function is analytic:

1) Use the definition of the derivative and the rules of differentiation to show that $f'(z)$ exists at every point of D.
2) Show that the first-order partial derivatives of u and v exist and are continuous in D (so u and v are differentiable) and satisfy the Cauchy–Riemann equations. Then $f(z)$ is analytic in D by Theorem 2.2.

2 Examples

Example 2.1

Determine where the following functions are differentiable and where they are analytic:

1. $w = \bar{z}$;
2. $f(z) = e^x(\cos y + i \sin y)$;
3. $w = z\mathrm{Re}(z)$.

2 Examples

Example 2.1

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2. $f(z) = e^x(\cos y + i \sin y)$;
3. $w = z\mathrm{Re}(z)$.

Solution.

1. $u = x$, $v = -y$: $\dfrac{\partial u}{\partial x} = 1$, $\dfrac{\partial v}{\partial y} = -1$. The Cauchy–Riemann equations never hold, so $w = \bar{z}$ is differentiable nowhere and analytic nowhere.

Solution (Cont.)

2 $u = e^x \cos y$, $v = e^x \sin y$, and the four partial derivatives

$$\frac{\partial u}{\partial x} = e^x \cos y, \quad \frac{\partial u}{\partial y} = -e^x \sin y,$$
$$\frac{\partial v}{\partial x} = e^x \sin y, \quad \frac{\partial v}{\partial y} = e^x \cos y$$

are continuous everywhere and satisfy $u_x = v_y$, $u_y = -v_x$. Therefore, $f(z)$ is analytic in the whole plane, and by (2.2.1)

$$f'(z) = u_x + iv_x = e^x(\cos y + i \sin y) = f(z).$$

This is the exponential function e^z of Section 3.

Solution (Cont.)

3 $w = z\mathrm{Re}(z) = x^2 + ixy$, so $u = x^2$, $v = xy$, and

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = 0, \quad \frac{\partial v}{\partial x} = y, \quad \frac{\partial v}{\partial y} = x$$

are continuous. The Cauchy–Riemann equations $2x = x$ and $0 = -y$ hold only at $x = y = 0$.
Hence $w = z\mathrm{Re}(z)$ is differentiable only at $z = 0$ and is analytic nowhere.

Example 2.2

Prove that $\overline{z}^2$ is analytic nowhere in the complex plane.

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Prove that $\bar{z}^2$ is analytic nowhere in the complex plane.

Proof.

$\bar{z}^2 = x^2 - y^2 - 2ixy$, so $u = x^2 - y^2$, $v = -2xy$, and

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial x} = -2y, \quad \frac{\partial v}{\partial y} = -2x.$$

The Cauchy–Riemann equations $2x = -2x$ and $-2y = 2y$ hold only at $x = y = 0$. Hence $\bar{z}^2$ is differentiable only at $z = 0$, and it is analytic nowhere. □

Example 2.3

Let $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$. *Find the constants* a, b, c, *and* d *such that* $f(z)$ *is analytic in the whole plane.*

Example 2.3

Let $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$. Find the constants a, b, c, and d such that $f(z)$ is analytic in the whole plane.

Solution.

$$\frac{\partial u}{\partial x} = 2x + ay, \quad \frac{\partial u}{\partial y} = ax + 2by, \quad \frac{\partial v}{\partial x} = 2cx + dy, \quad \frac{\partial v}{\partial y} = dx + 2y.$$

The Cauchy–Riemann equations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ require, for all x and y,

$$2x + ay = dx + 2y, \qquad ax + 2by = -2cx - dy.$$

Comparing coefficients gives $a = 2$, $b = -1$, $c = -1$, $d = 2$.

Example 2.4

Prove that $f(z) = \sqrt{|xy|}$ *satisfies the Cauchy–Riemann equations at* $z = 0$ *but is not differentiable at* $z = 0$.

Example 2.4

Prove that $f(z) = \sqrt{|xy|}$ satisfies the Cauchy–Riemann equations at $z = 0$ but is not differentiable at $z = 0$.

Proof.

Here $u = \sqrt{|xy|}$ and $v = 0$. At the origin,

$$u_x(0,0) = \lim_{x \to 0} \frac{u(x,0) - u(0,0)}{x} = 0 = v_y(0,0),$$

$$u_y(0,0) = \lim_{y \to 0} \frac{u(0,y) - u(0,0)}{y} = 0 = -v_x(0,0),$$

so the Cauchy–Riemann equations hold at $z = 0$.

Proof (Cont.)

However, as $z \to 0$ along the ray $y = kx$ $(x > 0,\ k > 0)$,

$$\frac{f(z) - f(0)}{z - 0} = \frac{\sqrt{k}\,x}{x + ikx} = \frac{\sqrt{k}}{1 + ik},$$

which depends on k. Therefore, $\lim_{z \to 0} \dfrac{f(z) - f(0)}{z}$ does not exist, and $f(z)$ is not differentiable at $z = 0$.

The Cauchy–Riemann equations alone are not sufficient: u must also be differentiable, and $\sqrt{|xy|}$ is not differentiable at $(0, 0)$. □

Example 2.5

Let $f(z) = u(x, y) + iv(x, y)$ be analytic in a domain D with $v = u^2$. Find $f(z)$.

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Solution.

By the Cauchy–Riemann equations,

$$u_x = v_y = 2uu_y, \tag{1}$$

$$u_y = -v_x = -2uu_x. \tag{2}$$

Substituting (2) into (1) gives $u_x(1 + 4u^2) = 0$. Since $1 + 4u^2 \neq 0$, $u_x = 0$, and then (2) gives $u_y = 0$.
Hence $u = c$ is a real constant in D, and $f(z) = c + ic^2$ is constant.

Example 2.6 (Classroom practice)

Let $f(z) = my^3 + nx^2y + i(x^3 + lxy^2)$ *be analytic. Determine* l, m, *and* n.

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Solution.

$u_x = 2nxy = v_y = 2lxy$ gives $n = l$;
$u_y = 3my^2 + nx^2 = -v_x = -3x^2 - ly^2$ gives $n = -3$ and $3m = -l$.
Hence

$$l = n = -3, \qquad m = 1.$$

Example 2.7

Suppose $f'(z) = 0$ everywhere in a domain D. Prove that $f(z)$ is constant in D.

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Proof.

By (2.2.1),

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y} \equiv 0,$$

so $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} \equiv 0$ in D. Since D is connected, u and v are constant, and hence $f(z)$ is constant in D. □

Example 2.8

Let $f(z) = u + iv$ be analytic with $f'(z) \neq 0$. Prove that the curve families $u(x, y) = c_1$ and $v(x, y) = c_2$ are orthogonal to each other, where c_1 and c_2 are constants.

Example 2.8

Let $f(z) = u + iv$ be analytic with $f'(z) \neq 0$. Prove that the curve families $u(x, y) = c_1$ and $v(x, y) = c_2$ are orthogonal to each other, where c_1 and c_2 are constants.

Proof.

By (2.2.1), $f'(z) = v_y - iu_y \neq 0$, so u_y and v_y are not both zero. First suppose $u_y \neq 0$ and $v_y \neq 0$ at an intersection point. By implicit differentiation, the slopes of the two curves are

$$k_1 = -\frac{u_x}{u_y}, \qquad k_2 = -\frac{v_x}{v_y}.$$

Proof (Cont.)

By the Cauchy–Riemann equations $u_x = v_y$ and $v_x = -u_y$,

$$k_1 \cdot k_2 = \left(-\frac{u_x}{u_y}\right) \cdot \left(-\frac{v_x}{v_y}\right) = \frac{v_y}{u_y} \cdot \frac{-u_y}{v_y} = -1,$$

so the two curves are orthogonal.
If one of u_y and v_y is zero, say $u_y = 0$ and $v_y \neq 0$, then $v_x = -u_y = 0$. Thus the tangent to $u = c_1$ is vertical and the tangent to $v = c_2$ is horizontal, and the curves are again orthogonal. □

Example 2.9

Prove that the real and imaginary parts of $f(z) = \sqrt{|\operatorname{Im} z^2|}$ *satisfy the Cauchy–Riemann equations at* $z = 0$*, but* $f(z)$ *is not differentiable at* $z = 0$*.*

Example 2.9

Prove that the real and imaginary parts of $f(z) = \sqrt{|\operatorname{Im} z^2|}$ *satisfy the Cauchy–Riemann equations at* $z = 0$*, but* $f(z)$ *is not differentiable at* $z = 0$*.*

Proof.

Since $\operatorname{Im} z^2 = 2xy$, we have $u = \sqrt{|2xy|}$ and $v = 0$. At the origin,

$$u_x(0,0) = \lim_{x \to 0} \frac{u(x,0) - u(0,0)}{x} = 0 = v_y(0,0),$$

$$u_y(0,0) = \lim_{y \to 0} \frac{u(0,y) - u(0,0)}{y} = 0 = -v_x(0,0),$$

so the Cauchy–Riemann equations hold at $z = 0$.

Proof (Cont.)

However,

$$\frac{f(\Delta z) - f(0)}{\Delta z} = \frac{\sqrt{|2\Delta x \Delta y|}}{\Delta x + i\Delta y}.$$

Along $\Delta y = \Delta x \to 0^+$, this quotient equals $\frac{\sqrt{2}}{1+i}$; along $\Delta x = 0$, $\Delta y \to 0$, it equals 0.
The two limits differ, so $f(z)$ is not differentiable at $z = 0$. □

3 Summary

A function $f(z) = u+iv$ is analytic in D if and only if $u(x, y)$ and $v(x, y)$ are differentiable in D and satisfy the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

In that case, $f'(z) = u_x + iv_x = v_y - iu_y$.

Example 2.10 (Question)

What should we check when using the Cauchy–Riemann equations to study the analyticity of $f(z) = u(x, y) + iv(x, y)$*?*

Solution.

1. Check whether $u(x, y)$ and $v(x, y)$ are differentiable in D (for example, whether their first-order partial derivatives are continuous);
2. Check whether they satisfy the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x};$$

3. Differentiability at isolated points or on a curve does not give analyticity: analyticity requires differentiability in a neighborhood.

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1 Exponential Function

1 Definition

For every complex number $z = x + iy$, the exponential function (指数函数) is defined by

$$\exp z = e^z = e^{x+iy} = e^x(\cos y + i \sin y).$$

- e^z is a notation for this function, not a power of the number e (powers are defined later).
- When $x = 0$, it reduces to Euler's formula $e^{iy} = \cos y + i \sin y$.
- When $y = 0$, it reduces to the real exponential function e^x.

2 Properties

1) $|e^z| = e^x$ and $\operatorname{Arg} e^z = y + 2k\pi$ $(k \in \mathbb{Z})$; in particular, $e^z \neq 0$.
2) $e^{z_1+z_2} = e^{z_1} e^{z_2}$ and $\dfrac{e^{z_1}}{e^{z_2}} = e^{z_1-z_2}$.

   However, $(e^{z_1})^{z_2} = e^{z_1 z_2}$ does not hold in general. For example, $(e^{i\pi})^{1/2} = (-1)^{1/2} = \pm i$, whereas $e^{i\pi/2} = i$.
3) e^z is analytic in the whole plane and $(e^z)' = e^z$.
4) e^z is periodic with fundamental period $2\pi i$: $e^{z+2k\pi i} = e^z$.

Example 3.1

Let $z = x + iy$. *Find*

1) $\left| e^{i-2z} \right|$;
2) $\left| e^{z^2} \right|$;
3) $\mathrm{Re}(e^{1/z})$.

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Let $z = x + iy$. *Find*

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3) $\mathrm{Re}(e^{1/z})$.

Solution.

Since $e^{x+iy} = e^x(\cos y + i \sin y)$, we have $\left|e^{x+iy}\right| = e^x$ and $\mathrm{Re}(e^{x+iy}) = e^x \cos y$.

Solution (Cont.)

1) $e^{i-2z} = e^{i-2(x+iy)} = e^{-2x+i(1-2y)}$, so $\left|e^{i-2z}\right| = e^{-2x}$.

2) $e^{z^2} = e^{(x+iy)^2} = e^{x^2-y^2+2ixy}$, so $\left|e^{z^2}\right| = e^{x^2-y^2}$.

3) $\dfrac{1}{z} = \dfrac{x-iy}{x^2+y^2}$, so $e^{1/z} = e^{\frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}}$ and

$$\operatorname{Re}\left(e^{1/z}\right) = e^{\frac{x}{x^2+y^2}} \cos \frac{y}{x^2+y^2}.$$

Example 3.2

Find the principal arguments of the following complex numbers:
(1) e^{2+i}; *(2)* e^{2-3i}; *(3)* e^{3+4i}; *(4)* e^{-3-4i};
(5) $e^{i\alpha} - e^{i\beta} \quad (0 \leq \beta < \alpha \leq 2\pi)$.

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(5) $e^{i\alpha} - e^{i\beta} \quad (0 \le \beta < \alpha \le 2\pi)$.

Solution.

Since $\operatorname{Arg} e^{x+iy} = y + 2k\pi \ (k \in \mathbb{Z})$, the principal argument $\arg e^{x+iy}$ is the value of $y + 2k\pi$ in $(-\pi, \pi]$.

1. $\operatorname{Arg} e^{2+i} = 1 + 2k\pi$, so $\arg e^{2+i} = 1$;
2. $\operatorname{Arg} e^{2-3i} = -3 + 2k\pi$, so $\arg e^{2-3i} = -3$;

Solution (Cont.)

(3) $\operatorname{Arg} e^{3+4i} = 4 + 2k\pi$; since $4 > \pi$, $\arg e^{3+4i} = 4 - 2\pi$;

(4) $\operatorname{Arg} e^{-3-4i} = -4+2k\pi$; since $-4 < -\pi$, $\arg e^{-3-4i} = -4+2\pi$;

(5)
$$\begin{aligned} e^{i\alpha} - e^{i\beta} &= (\cos\alpha - \cos\beta) + i(\sin\alpha - \sin\beta) \\ &= -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} + 2i\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} \\ &= 2\sin\frac{\alpha-\beta}{2}\left(-\sin\frac{\alpha+\beta}{2} + i\cos\frac{\alpha+\beta}{2}\right) \\ &= 2\sin\frac{\alpha-\beta}{2}\left(\cos\frac{\pi+\alpha+\beta}{2} + i\sin\frac{\pi+\alpha+\beta}{2}\right). \end{aligned}$$

Solution (Cont.)

Since $0 \le \beta < \alpha \le 2\pi$, we have $\sin \dfrac{\alpha - \beta}{2} > 0$, so the last expression is the trigonometric form of $e^{i\alpha} - e^{i\beta}$. Hence

$$\operatorname{Arg}(e^{i\alpha} - e^{i\beta}) = \frac{\pi + \alpha + \beta}{2} + 2k\pi,$$

and

$$\arg(e^{i\alpha} - e^{i\beta}) = \begin{cases} \dfrac{\pi + \alpha + \beta}{2}, & \alpha + \beta \le \pi, \\ \dfrac{\pi + \alpha + \beta}{2} - 2\pi, & \alpha + \beta > \pi. \end{cases}$$

Example 3.3

Find the period of $f(z) = e^{z/5}$.

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Find the period of $f(z) = e^{z/5}$.

Solution.

Since e^w has periods $2k\pi i$ $(k \in \mathbb{Z})$,

$$f(z) = e^{\frac{z}{5}} = e^{\frac{z}{5}+2k\pi i} = e^{\frac{z+10k\pi i}{5}} = f(z+10k\pi i).$$

Hence the periods of $f(z)$ are $10k\pi i$ $(k \neq 0)$, and its fundamental period is $10\pi i$.

2 Logarithmic Function

1. Definition
   A function $w = f(z)$ satisfying $e^w = z$ $(z \neq 0)$ is called the logarithmic function (对数函数) and is denoted by $w = \operatorname{Ln} z$.
2. Let $w = u + iv$ and $z = re^{i\theta}$. Then

$$e^w = z \iff e^u e^{iv} = re^{i\theta} \iff \begin{cases} e^u = r, \\ v = \theta + 2k\pi \ (k \in \mathbb{Z}), \end{cases}$$

so $u = \ln r$ and $v = \theta + 2k\pi$. Therefore,

$$\operatorname{Ln} z = \ln |z| + i \operatorname{Arg} z.$$

Since $\operatorname{Arg} z$ is multiple-valued, so is $\operatorname{Ln} z$.

If $\operatorname{Arg} z$ is replaced by the principal argument $\arg z$, we obtain a single-valued function, the principal value（主値）of $\operatorname{Ln} z$:

$$\ln z = \ln |z| + i \arg z.$$

All the values are

$$\operatorname{Ln} z = \ln z + 2k\pi i \quad (k \in \mathbb{Z}).$$

For each fixed k, this formula defines a single-valued function called a branch（分支）of $\operatorname{Ln} z$.
When $z = x > 0$, the principal value is $\ln z = \ln x$, the real logarithm.

Example 3.4

Find $\operatorname{Ln} 2$ *and* $\operatorname{Ln}(-1)$ *and their principal values.*

Example 3.4

Find $\operatorname{Ln} 2$ *and* $\operatorname{Ln}(-1)$ *and their principal values.*

Solution.

$\operatorname{Ln} 2 = \ln 2 + 2k\pi i \ (k \in \mathbb{Z})$, and its principal value is $\ln 2$.
$\operatorname{Ln}(-1) = \ln 1 + i\operatorname{Arg}(-1) = (2k+1)\pi i \ (k \in \mathbb{Z})$, and its principal value is πi.

Note: A negative real number has no real logarithm, but it has complex logarithms. The complex logarithm extends the real logarithm.

Example 3.5

Solve the equation $e^z - 1 - \sqrt{3}i = 0$.

Example 3.5

Solve the equation $e^z - 1 - \sqrt{3}i = 0$.

Solution.

Since $e^z = 1 + \sqrt{3}i$,

$$z = \operatorname{Ln}(1 + \sqrt{3}i) = \ln\left|1 + \sqrt{3}i\right| + i\operatorname{Arg}(1 + \sqrt{3}i)$$
$$= \ln 2 + i\left(\frac{\pi}{3} + 2k\pi\right) \quad (k \in \mathbb{Z}).$$

Example 3.6

Find (1) $\mathrm{Ln}(-2+3i)$*; (2)* $\mathrm{Ln}(3-\sqrt{3}i)$*; (3)* $\mathrm{Ln}(-3)$.

Example 3.6

Find (1) $\mathrm{Ln}(-2+3i)$*; (2)* $\mathrm{Ln}(3-\sqrt{3}i)$*; (3)* $\mathrm{Ln}(-3)$.

Solution.

(1) $-2+3i$ lies in the second quadrant, so $\arg(-2+3i) = \pi - \arctan\frac{3}{2}$ and

$$\mathrm{Ln}(-2+3i) = \frac{1}{2}\ln 13 + i\left(\pi - \arctan\frac{3}{2} + 2k\pi\right) \quad (k \in \mathbb{Z}).$$

Solution (Cont.)

(2) $\left|3-\sqrt{3}i\right| = 2\sqrt{3}$ and $\arg(3-\sqrt{3}i) = \arctan\dfrac{-\sqrt{3}}{3} = -\dfrac{\pi}{6}$, so

$$\operatorname{Ln}(3-\sqrt{3}i) = \ln(2\sqrt{3}) + i\left(2k\pi - \frac{\pi}{6}\right) \quad (k\in\mathbb{Z}).$$

(3) $|-3| = 3$ and $\arg(-3) = \pi$, so

$$\operatorname{Ln}(-3) = \ln 3 + (2k+1)\pi i \quad (k\in\mathbb{Z}).$$

3 Properties

1) $\operatorname{Ln}(z_1 z_2) = \operatorname{Ln} z_1 + \operatorname{Ln} z_2$;
2) $\operatorname{Ln} \dfrac{z_1}{z_2} = \operatorname{Ln} z_1 - \operatorname{Ln} z_2$.
   These are equalities of sets, like $\operatorname{Arg}(z_1 z_2) = \operatorname{Arg} z_1 + \operatorname{Arg} z_2$; for principal values, $\ln(z_1 z_2) \neq \ln z_1 + \ln z_2$ in general.
3) In the plane with the origin and the negative real axis removed, the principal branch and every other branch are continuous and differentiable, and

$$(\ln z)' = \frac{1}{z}, \qquad (\operatorname{Ln} z)' = \frac{1}{z}.$$

Proof of (3).

Let $z = x + iy$. For $x < 0$,

$$\lim_{y \to 0^-} \arg z = -\pi, \qquad \lim_{y \to 0^+} \arg z = \pi,$$

so $\arg z$, and hence $\ln z$, is discontinuous on the negative real axis. Elsewhere, except at the origin, $\ln z$ is continuous.
In the region $-\pi < \arg z < \pi$, $w = \ln z$ is the single-valued inverse of $z = e^w$, so

$$\frac{\mathrm{d} \ln z}{\mathrm{d} z} = \frac{1}{\frac{\mathrm{d} e^w}{\mathrm{d} w}} = \frac{1}{e^w} = \frac{1}{z}.$$



3 Powers a^b and Power Functions

1 Definition of a^b

Let $a \neq 0$ and let b be any complex number. The power (乗冪) a^b is defined by

$$a^b = e^{b \operatorname{Ln} a}.$$

Since $\operatorname{Ln} a = \ln|a| + i(\arg a + 2k\pi)$ is multiple-valued, a^b is multiple-valued in general.

1) If b is an integer, then $e^{2kb\pi i} = 1$ and

$$a^b = e^{b[\ln|a| + i(\arg a + 2k\pi)]} = e^{b \ln a}\, e^{2kb\pi i} = e^{b \ln a},$$

so a^b is single-valued.

2) If $b = \dfrac{p}{q}$, where p and q are coprime integers and $q > 0$, then

$$\begin{aligned} a^b &= e^{\frac{p}{q}\ln|a| + i\frac{p}{q}(\arg a + 2k\pi)} \\ &= e^{\frac{p}{q}\ln|a|}\left[\cos\frac{p}{q}(\arg a + 2k\pi) + i\sin\frac{p}{q}(\arg a + 2k\pi)\right]. \end{aligned}$$

It has exactly q distinct values, obtained for $k = 0, 1, \ldots, q-1$.

3) If b is irrational or non-real, a^b has infinitely many values.

Special cases:

1) If $b = n$ is a positive integer,

$$a^n = e^{n \ln a} = \left(e^{\ln a}\right)^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}},$$

which agrees with the power defined in Chapter 1.

2) If $b = \dfrac{1}{n}$,

$$a^{\frac{1}{n}} = |a|^{\frac{1}{n}} \left[\cos \frac{\arg a + 2k\pi}{n} + i \sin \frac{\arg a + 2k\pi}{n}\right] = \sqrt[n]{a},$$

$k = 0, 1, \ldots, n-1$, which are exactly the n values of the n-th root in (1.2.10).

If $a = z$ is a complex variable, we obtain the power function (冪函数) $w = z^b$.

For $b = n$ and $b = \dfrac{1}{n}$, we obtain $w = z^n$ and its inverse, the n-th root function $w = z^{\frac{1}{n}} = \sqrt[n]{z}$.

Example 3.7

Find the values of $1^{\sqrt{2}}$ *and* i^i.

Example 3.7

Find the values of $1^{\sqrt{2}}$ *and* i^i.

Solution.

$$1^{\sqrt{2}} = e^{\sqrt{2}\,\mathrm{Ln}\,1} = e^{2\sqrt{2}k\pi i} = \cos(2\sqrt{2}k\pi) + i\sin(2\sqrt{2}k\pi) \quad (k \in \mathbb{Z}),$$

$$i^i = e^{i\,\mathrm{Ln}\,i} = e^{i\left(\frac{\pi}{2}+2k\pi\right)i} = e^{-\left(\frac{\pi}{2}+2k\pi\right)} \quad (k \in \mathbb{Z}).$$

All values of i^i are positive real numbers.

Example 3.8 (Classroom practice)

Find the values of $(-3)^{\sqrt{5}}$.

Example 3.8 (Classroom practice)

Find the values of $(-3)^{\sqrt{5}}$.

Solution.

$$
\begin{aligned}
(-3)^{\sqrt{5}} &= e^{\sqrt{5}\,\mathrm{Ln}(-3)} = e^{\sqrt{5}[\ln 3+(2k+1)\pi i]} \\
&= 3^{\sqrt{5}}\left[\cos\sqrt{5}(2k+1)\pi + i\sin\sqrt{5}(2k+1)\pi\right] \quad (k\in\mathbb{Z}).
\end{aligned}
$$

Example 3.9

Find the principal argument of $(1+i)^i$.

Example 3.9

Find the principal argument of $(1+i)^i$.

Solution.

$$(1+i)^i = e^{i\,\mathrm{Ln}(1+i)} = e^{i\left[\frac{1}{2}\ln 2 + i\left(\frac{\pi}{4}+2k\pi\right)\right]}$$
$$= e^{-\left(\frac{\pi}{4}+2k\pi\right)}\left[\cos\left(\frac{1}{2}\ln 2\right) + i\sin\left(\frac{1}{2}\ln 2\right)\right] \quad (k\in\mathbb{Z}).$$

All values have modulus $e^{-\left(\frac{\pi}{4}+2k\pi\right)} > 0$ and the same argument $\frac{1}{2}\ln 2 \in (-\pi,\pi]$. Therefore, the principal argument of $(1+i)^i$ is $\frac{1}{2}\ln 2$.

2 Analyticity of Power Functions

1) z^n is single-valued and analytic in the whole plane, and $(z^n)' = nz^{n-1}$.
2) $z^{\frac{1}{n}} = e^{\frac{1}{n}\operatorname{Ln} z}$ is multiple-valued with n branches. Each branch is analytic in the plane with the origin and the negative real axis removed, and
$$\left(z^{\frac{1}{n}}\right)' = \left(e^{\frac{1}{n}\operatorname{Ln} z}\right)' = \frac{1}{n} z^{\frac{1}{n}-1}.$$
3) For other exponents b, $z^b = e^{b\operatorname{Ln} z}$ is multiple-valued (with infinitely many branches if b is irrational or non-real). Each branch is analytic in the same cut plane, and
$$\left(z^b\right)' = bz^{b-1}.$$

4 Trigonometric and Hyperbolic Functions

1 Definition of Trigonometric Functions

For real y, $e^{iy} = \cos y + i \sin y$ and $e^{-iy} = \cos y - i \sin y$. Adding and subtracting gives

$$\cos y = \frac{e^{iy} + e^{-iy}}{2}, \qquad \sin y = \frac{e^{iy} - e^{-iy}}{2i}.$$

Replacing the real number y by a complex number z extends cosine and sine to the complex plane.

The cosine and sine functions (余弦、正弦函数) are defined by

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

$\sin z$ is odd and $\cos z$ is even:

$$\sin(-z) = -\sin z, \qquad \cos(-z) = \cos z.$$

Both are periodic with fundamental period 2π:

$$\sin(z + 2\pi) = \sin z, \qquad \cos(z + 2\pi) = \cos z.$$

Example 3.10

Find the period of $f(z) = \sin 5z$.

Example 3.10

Find the period of $f(z) = \sin 5z$.

Solution.

Since $\sin w$ has period 2π,

$$f\left(z + \frac{2\pi}{5}\right) = \sin(5z + 2\pi) = \sin 5z = f(z).$$

Therefore, the fundamental period of $f(z) = \sin 5z$ is $\dfrac{2\pi}{5}$.

Sine and cosine are analytic in the whole plane, and

$$(\sin z)' = \cos z, \qquad (\cos z)' = -\sin z.$$

Important formulas:

(1
$$\begin{cases} \cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2, \\ \sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2, \\ \sin^2 z + \cos^2 z = 1; \end{cases}$$

(2
$$\begin{cases} \cos(x + iy) = \cos x \cos iy - \sin x \sin iy, \\ \sin(x + iy) = \sin x \cos iy + \cos x \sin iy. \end{cases}$$

(3) For a purely imaginary number iy,

$$\cos iy = \frac{e^{-y} + e^{y}}{2} = \cosh y, \qquad \sin iy = \frac{e^{-y} - e^{y}}{2i} = i \sinh y.$$

(4) Hence

$$\begin{cases} \cos(x + iy) = \cos x \cosh y - i \sin x \sinh y, \\ \sin(x + iy) = \sin x \cosh y + i \cos x \sinh y. \end{cases}$$

As $y \to \infty$, $|\sin iy| \to \infty$ and $|\cos iy| \to \infty$. Unlike the real case, $\sin z$ and $\cos z$ are unbounded.

The other trigonometric functions are defined by

$$\tan z = \frac{\sin z}{\cos z}, \quad \cot z = \frac{\cos z}{\sin z}, \quad \sec z = \frac{1}{\cos z}, \quad \csc z = \frac{1}{\sin z}.$$

Their periodicity, parity, and analyticity can be studied in the same way as for $\sin z$ and $\cos z$.

Example 3.11

Find the real and imaginary parts of $\tan z$.

Example 3.11

Find the real and imaginary parts of $\tan z$.

Solution.

Let $z = x + iy$. Then

$$\tan z = \frac{\sin(x+iy)}{\cos(x+iy)} = \frac{\sin x \cosh y + i \cos x \sinh y}{\cos x \cosh y - i \sin x \sinh y}.$$

Solution (Cont.)

Multiplying numerator and denominator by the conjugate of the denominator,

$$\begin{aligned}\tan z &= \frac{\sin x\cos x + i\sinh y\cosh y}{\cos^2 x\cosh^2 y + \sin^2 x\sinh^2 y} \\ &= \frac{\sin 2x + i\sinh 2y}{\cos 2x + \cosh 2y},\end{aligned}$$

using

$$\cos^2 x\cosh^2 y + \sin^2 x\sinh^2 y = \cos^2 x + \sinh^2 y = \frac{\cos 2x + \cosh 2y}{2}.$$

Hence

$$\mathrm{Re}(\tan z) = \frac{\sin 2x}{\cos 2x + \cosh 2y}, \qquad \mathrm{Im}(\tan z) = \frac{\sinh 2y}{\cos 2x + \cosh 2y}.$$

Example 3.12

Solve the equation $\sin z = i \sinh 1$.

Example 3.12

Solve the equation $\sin z = i\sinh 1$.

Solution.

Let $z = x + iy$. Then $\sin x\cosh y + i\cos x\sinh y = i\sinh 1$, so

$$\sin x\cosh y = 0, \qquad \cos x\sinh y = \sinh 1.$$

Since $\cosh y \geq 1$, $\sin x = 0$ and $x = k\pi$. Then $\sinh y = (-1)^k \sinh 1$, so $y = (-1)^k$. Therefore,

$$z = \begin{cases} 2n\pi + i, \\ (2n+1)\pi - i, \end{cases} \qquad n \in \mathbb{Z}.$$

Example 3.13

Find the values of $\cos(1+i)$ *and* $\tan(3-i)$.

Example 3.13

Find the values of $\cos(1+i)$ *and* $\tan(3-i)$.

Solution.

$$\begin{aligned}
\cos(1+i) &= \frac{e^{i(1+i)} + e^{-i(1+i)}}{2} = \frac{e^{-1+i} + e^{1-i}}{2} \\
&= \frac{1}{2}\left[e^{-1}(\cos 1 + i\sin 1) + e(\cos 1 - i\sin 1)\right] \\
&= \frac{e+e^{-1}}{2}\cos 1 - i\frac{e-e^{-1}}{2}\sin 1 \\
&= \cos 1 \cosh 1 - i \sin 1 \sinh 1.
\end{aligned}$$

Solution (Cont.)

By the formula for $\tan z$ with $x = 3$ and $y = -1$,

$$\tan(3 - i) = \frac{\sin 6 + i \sinh(-2)}{\cos 6 + \cosh(-2)} = \frac{\sin 6 - i \sinh 2}{\cos 6 + \cosh 2}.$$

Directly, $\sin(3 - i) = \sin 3 \cosh 1 - i \cos 3 \sinh 1$ and $\cos(3 - i) = \cos 3 \cosh 1 + i \sin 3 \sinh 1$; multiplying by the conjugate of the denominator gives the same result.

2 Definition of Hyperbolic Functions
The hyperbolic cosine, sine, and tangent functions（双曲函数）are defined by

$$\cosh z = \frac{e^z + e^{-z}}{2}, \qquad \sinh z = \frac{e^z - e^{-z}}{2},$$

$$\tanh z = \frac{\sinh z}{\cosh z} = \frac{e^z - e^{-z}}{e^z + e^{-z}}.$$

For real z, they agree with the real hyperbolic functions from calculus.

$\sinh z$ is odd and $\cosh z$ is even. Both are periodic with fundamental period $2\pi i$, and

$$(\sinh z)' = \cosh z, \qquad (\cosh z)' = \sinh z.$$

Moreover,

$$\cosh iy = \cos y, \qquad \sinh iy = i\sin y,$$

$$\begin{cases} \cosh(x+iy) = \cosh x\cos y + i\sinh x\sin y, \\ \sinh(x+iy) = \sinh x\cos y + i\cosh x\sin y. \end{cases}$$

Example 3.14

Solve the equation $|\tanh z| = 1$.

Example 3.14

Solve the equation $|\tanh z| = 1$.

Solution.

$\tanh z = \dfrac{e^{2z}-1}{e^{2z}+1}$, so $|\tanh z| = 1$ means $\left|e^{2z}-1\right| = \left|e^{2z}+1\right|$. Let $e^{2z} = u + iv$. Squaring gives

$$(u-1)^2 + v^2 = (u+1)^2 + v^2 \iff u = 0.$$

Since $u = \mathrm{Re}(e^{2z}) = e^{2x}\cos 2y$,

$$u = 0 \iff \cos 2y = 0 \iff y = \frac{\pi}{4} + \frac{k\pi}{2} \quad (k \in \mathbb{Z}).$$

Hence the solutions are all z with $\mathrm{Im}\, z = \dfrac{\pi}{4} + \dfrac{k\pi}{2}$ $(k \in \mathbb{Z})$: a family of horizontal lines.

5 Summary

The complex elementary functions extend the real elementary functions. They keep many basic properties, but some new features appear:

1. The exponential function is periodic, with period $2\pi i$;
2. Negative numbers have logarithms, and $\operatorname{Ln} z$ is multiple-valued;
3. $\sin z$ and $\cos z$ are unbounded;
4. The hyperbolic sine and cosine are periodic, with period $2\pi i$.

Example 3.15 (Question)

What are the similarities and differences between real and complex trigonometric functions?

Solution.

- Similarities: parity, periodicity, differentiability, the formulas for derivatives, the addition formulas, and $\sin^2 z + \cos^2 z = 1$.
- Main difference: real sine and cosine are bounded by 1, but $|\sin z| \leq 1$ and $|\cos z| \leq 1$ fail for complex z. For example, $|\cos iy| = \cosh y \to \infty$.